

4. Data & Database Documentation

Database Schema (overview)

- Primary stores: relational database (Postgres) — tables: users, bookings, slots, payments, audit_logs.
- Caching: Redis for session/counters.

ER Diagram

- See `diagrams/er_diagram.mmd` for a mermaid ER starter.

Data Dictionary (example)

- bookings:
 - `id` (uuid): PK
 - `user_id` (uuid): FK -> users.id
 - `slot_id` (uuid): FK -> slots.id
 - `status` (varchar): enum(created, confirmed, cancelled)
 - `created_at` (timestamp)

Relationships

- One `user` to many `bookings`.

Indexes & Performance

- Indexes: `bookings(user_id)`, `bookings(slot_id)`, `bookings(created_at)` — tune for read patterns.

Data Lifecycle & Retention

- Raw retention: 90 days in primary DB, archival to object storage for 7 years (configurable).
- Deletion: soft-delete + GDPR purge endpoint that removes PII within configured SLA.

Backups & Recovery

- Daily DB snapshots, WAL archiving for point-in-time recovery.